

Задание 1

Программа:

```
program pr1;
var
  a, lam, h, y, r: real;
  Q, Qn, Qk: real;
begin
  writeln('Введите значения h, a, lam:');
  readln(h, a, lam);
  writeln('Введите значения Qнач и Qкон:');
  readln(Qn, Qk);
  Q := Qn;
  Repeat
    r := (Pi * (a / lam)*cos(Q));
    y := (1 + sin(Q)) * cos(r)/((Pi/2)*(Pi/2)- r*r);
    writeln('Q = ', Q);
    writeln('y = ',y);
    Q := Q + h;
  until Q>=Qk;
end.
```

Результат:

Окно вывода	
5	
13.5	
3	
Введите значения Qнач и Qкон:	
0	
90	
Q = 0	
y = -2.79176771846724E-18	
Q = 5	
y = 0.00194878655658887	
Q = 10	Q = 50
y = -0.00251365603258225	y = -0.00190911492471805
Q = 15	Q = 55
y = 0.00369835781402023	y = 9.83080105097564E-05
Q = 20	Q = 60
y = -0.0540543561329012	y = -0.0024224890763558
Q = 25	Q = 65
y = -0.000555100779049746	y = 0.00292753888194686
Q = 30	Q = 70
y = 0.00299618189708408	y = 0.0203409438686986
Q = 35	Q = 75
y = -0.00347957617486593	y = -0.00327079320705338
Q = 40	Q = 80
y = 0.0201905885258496	y = 0.00195162515031931
Q = 45	Q = 85
y = -0.0145623629706637	y = -0.000944079648127343